



LECTURE 3

DEEP LEARNING I

What is Learning?

PRESS → OR SPACE. →

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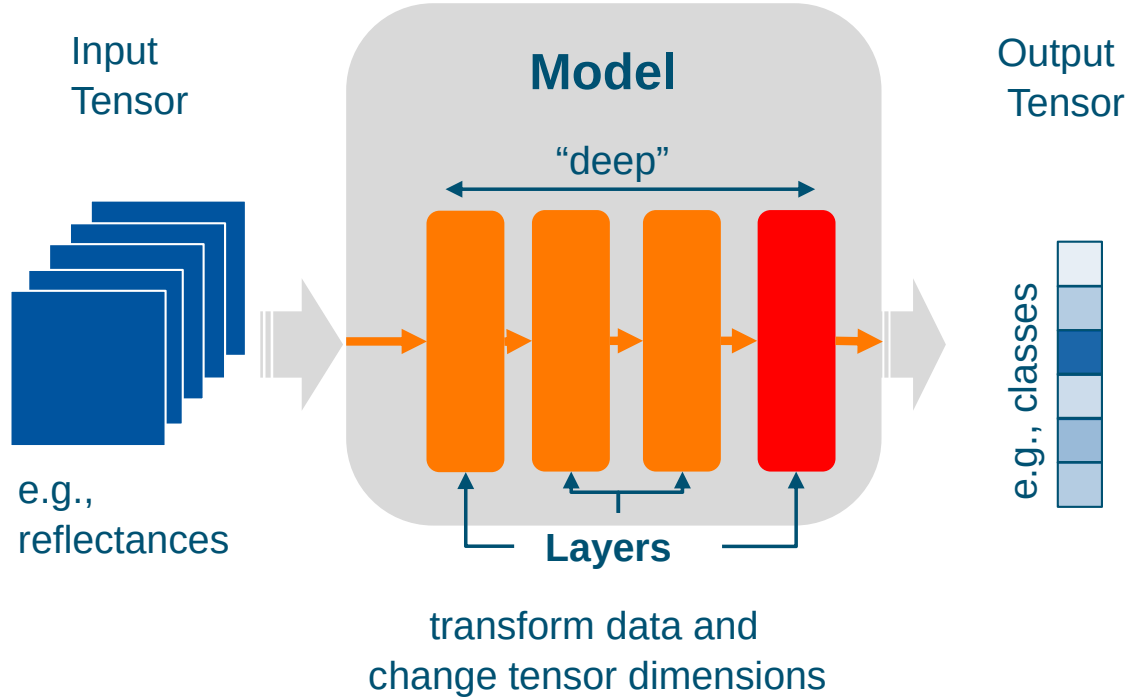
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In this lecture

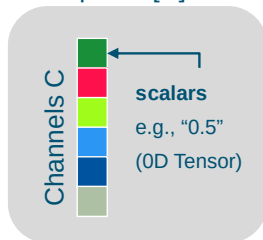
- Inputs, targets, and supervised learning
- Classification and regression
- Train / validation / test splits
- Decision trees and ensemble methods
- Random forests as the main model

Model

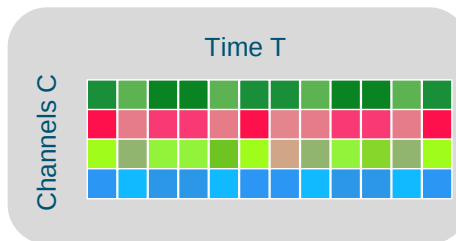


Tensors

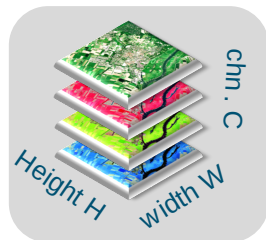
1D Tensor: Vector
e.g., pixels or
class probs [C]



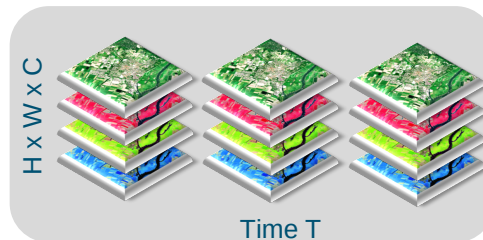
2D Tensor: Matrix
Time Series [T x C]



3D Tensors
Images [H x W x C]



4D Tensors
Img. Time Series [T x H x W x C]



Sentinel-2 as an Image Tensor

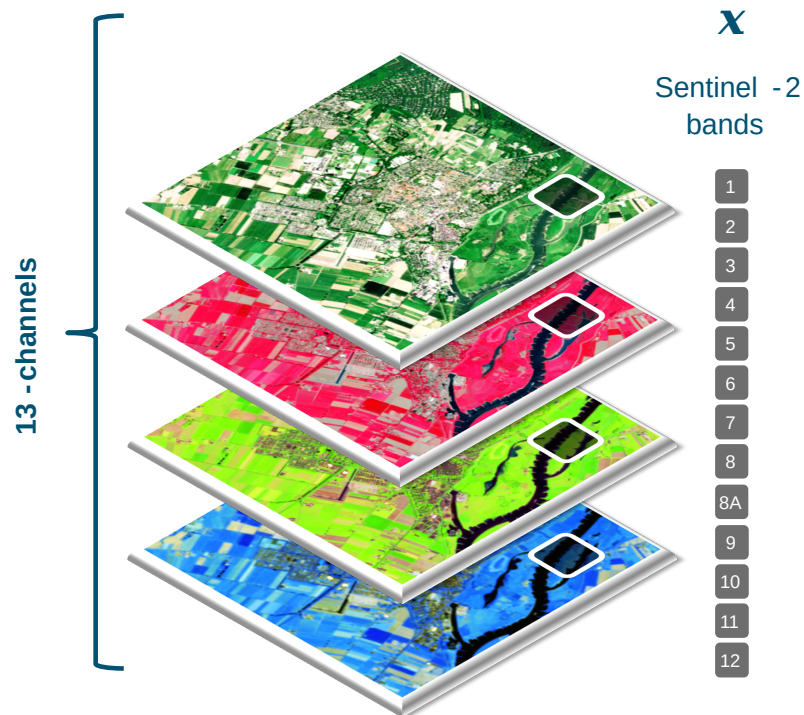


Image Classification

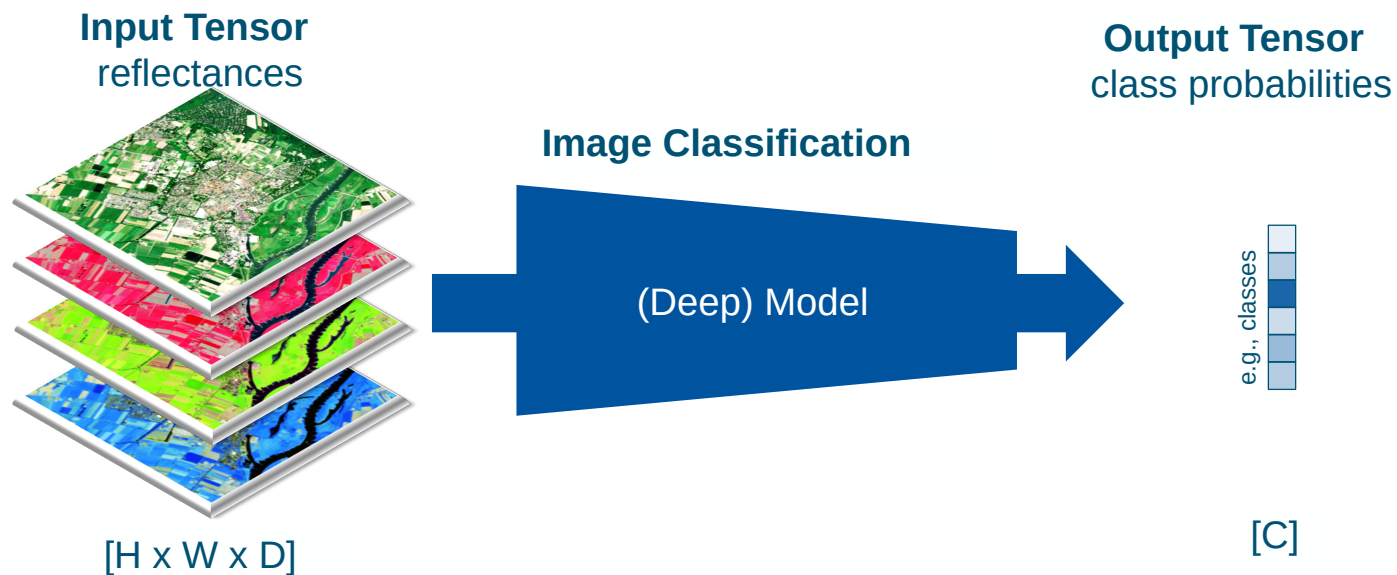
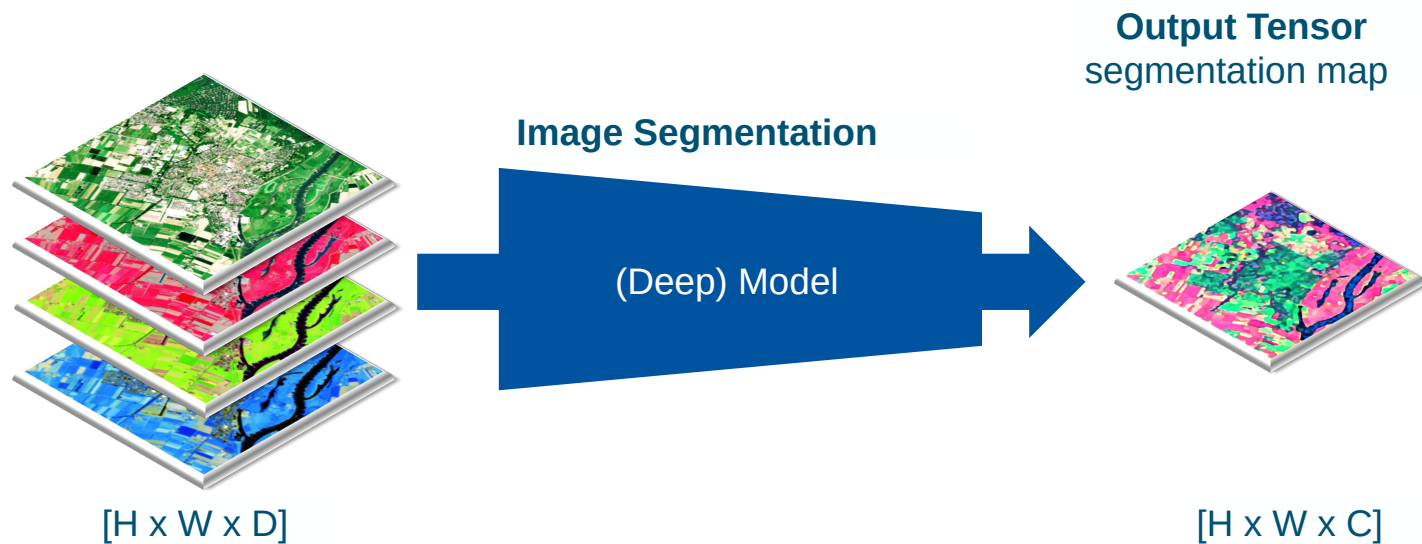


Image Segmentation



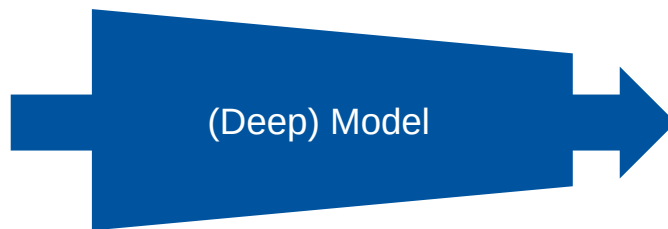
Object Detection

Input Tensor
reflectances



[H x W x 3]

Object Detection

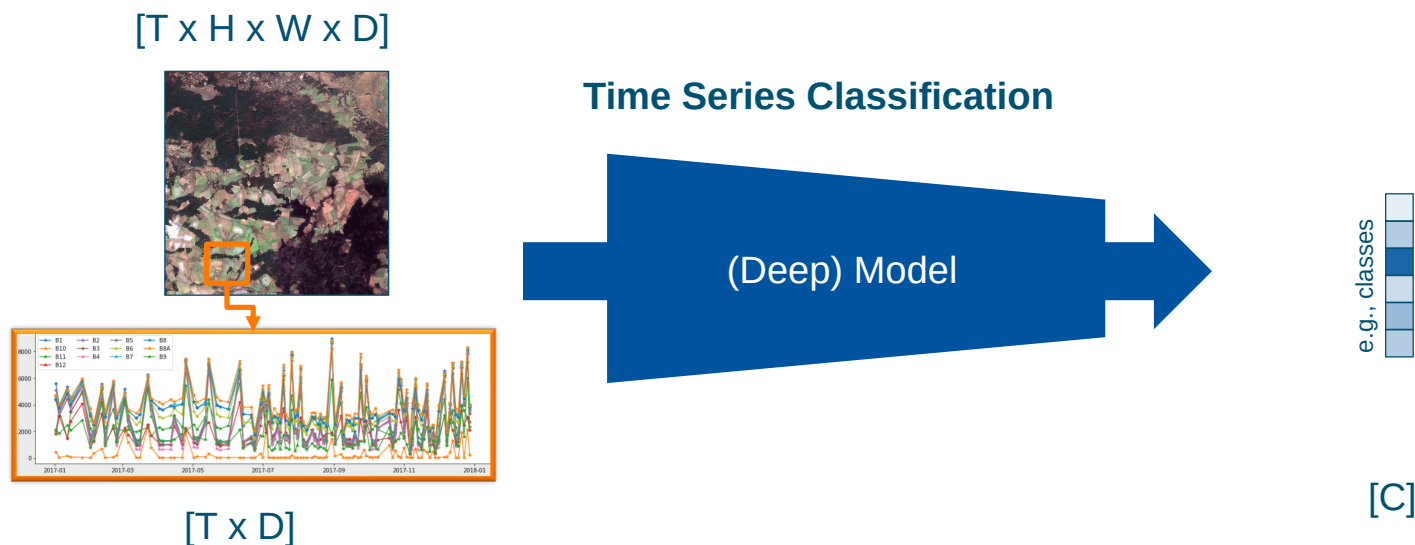


Output Tensor
bounding boxes

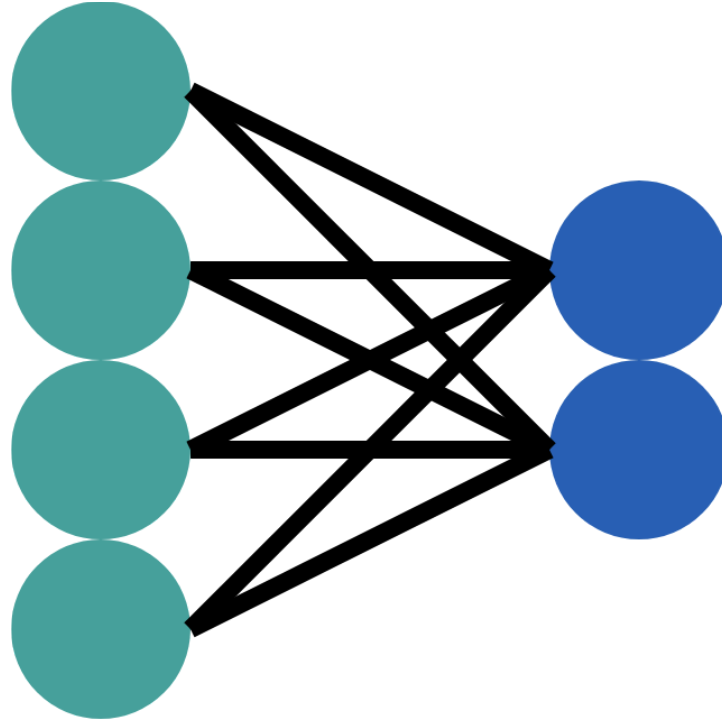
X	Y	H	W	
Green	Green	Green	Green	Bbox 1
Red	Red	Red	Red	Bbox 2
Yellow	Yellow	Yellow	Yellow	Bbox 3
Blue	Blue	Blue	Blue	...

[N x 4]

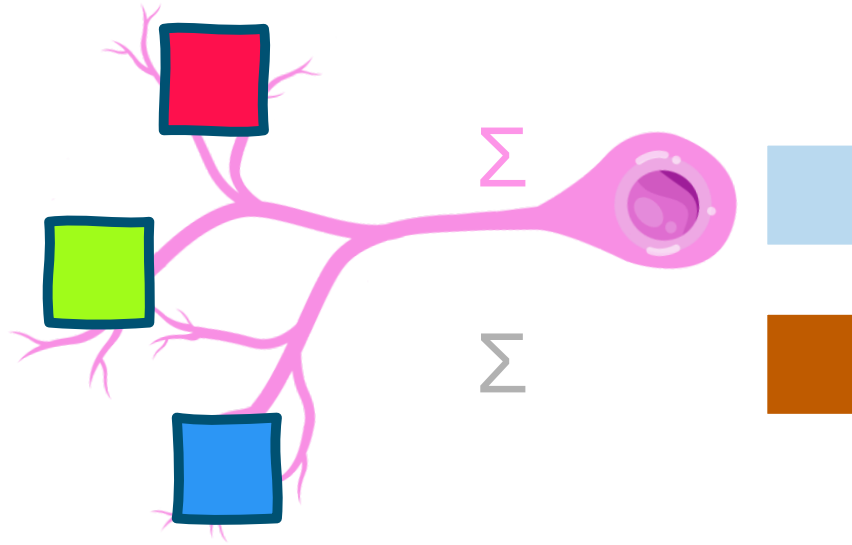
Time Series Classification



Multilayer Perceptron



Biological Interpretation

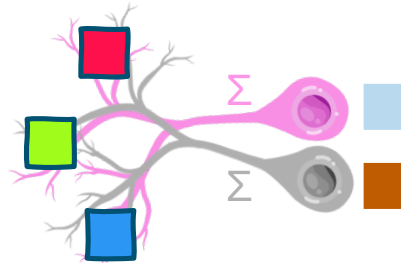


Neuron Inputs and Outputs

Biological Interpretation

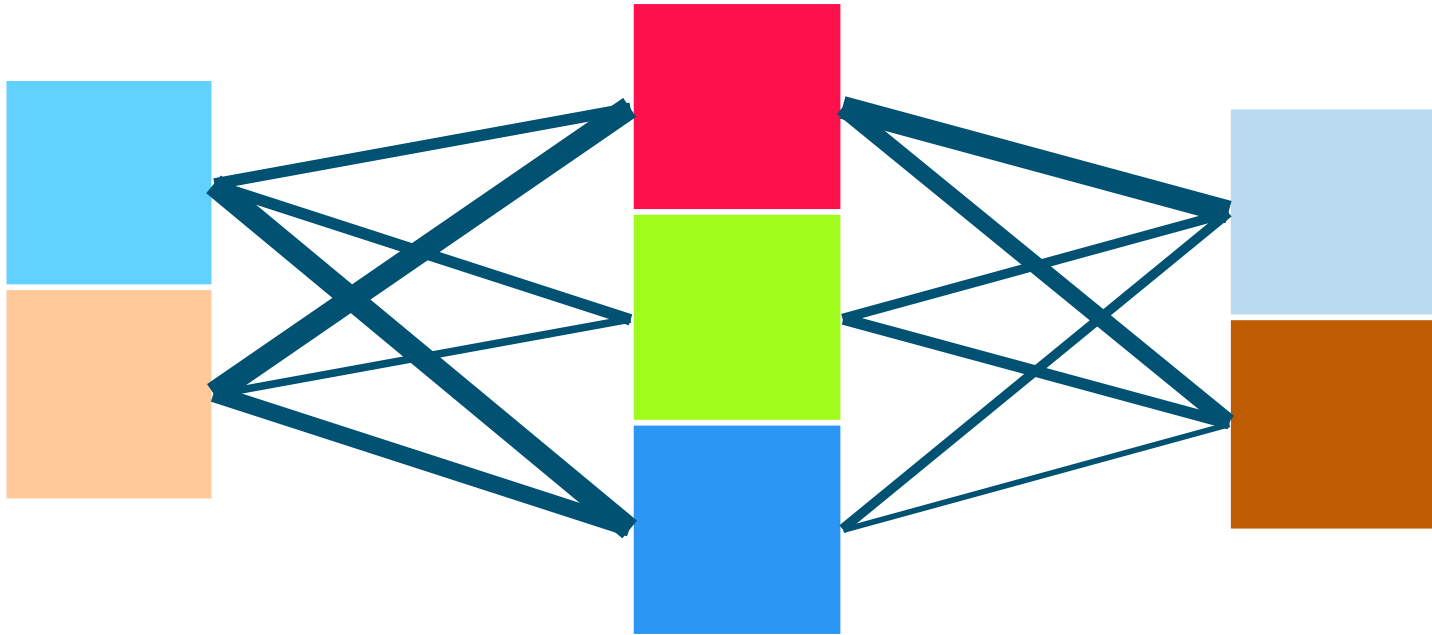
Inputs

Outputs

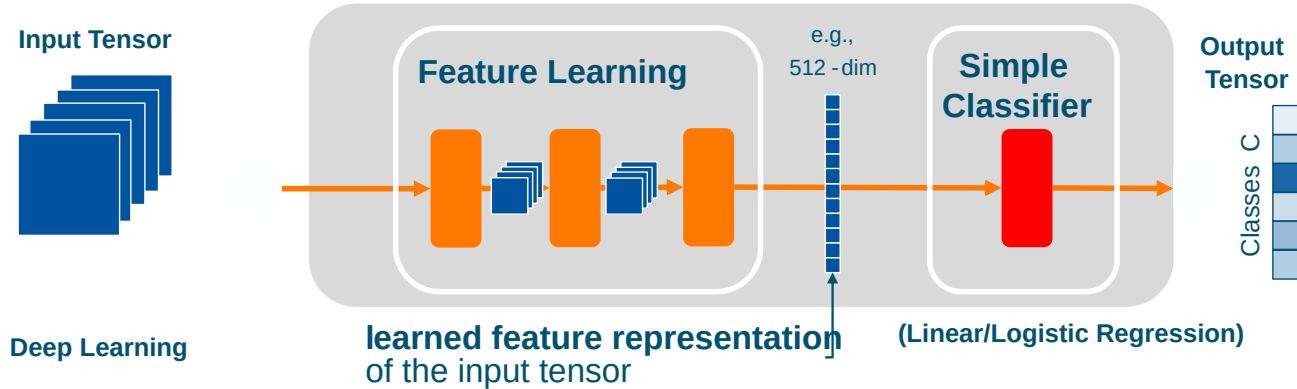


Neuron Image: [Wikipedia](#) by [Hariadhi](#) - Own work

Weighted Connections



Machine Learning and Deep Learning



Machine Learning

Input Tensor



Design knowledge in **designing** features mathematically

Features Design

(e.g., NDVI, NDWI, GLCM Textures)

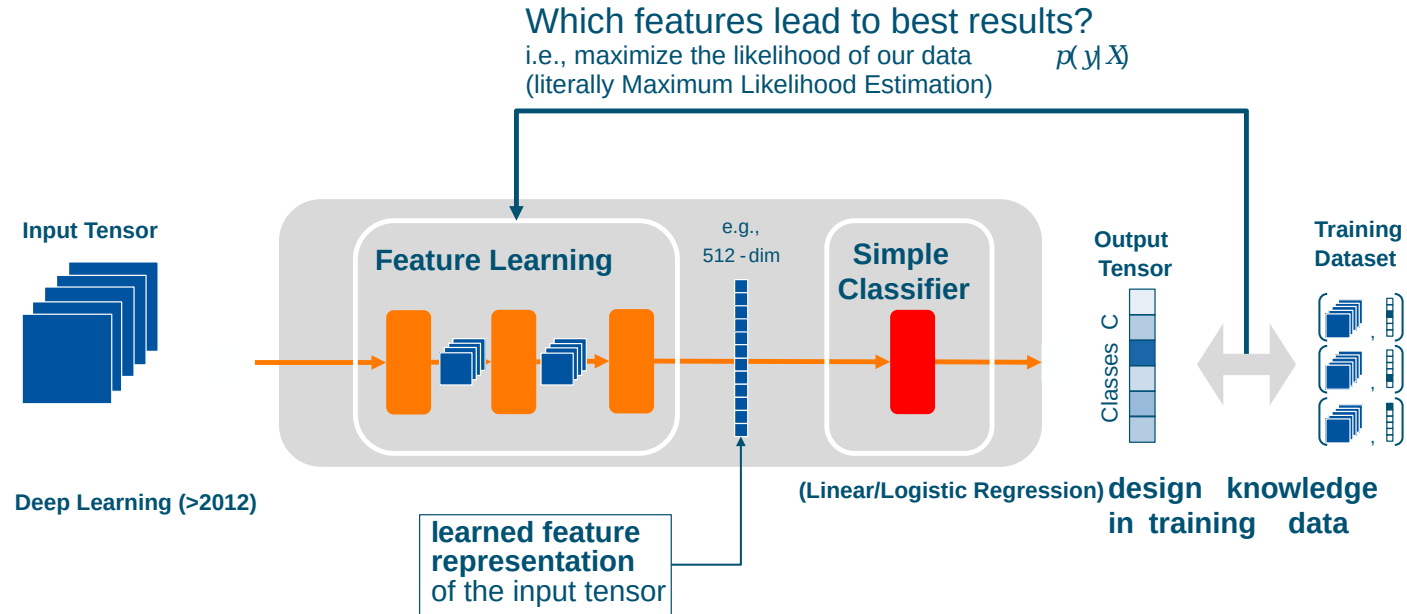
Complex Classifier

(Random Forest, Boosting)

Output Tensor



Feature Learning



Foundation Models

